Layer-Adaptive Visual Reinforcement for Vision-Language Navigation

Anonymous submission

Abstract

Vision-Language Navigation (VLN) requires embodied agents to navigate complex environments by grounding natural-language instructions in visual observations. However, effectively leveraging visual information throughout reasoning remains challenging. Our empirical analysis reveals that visual information is progressively attenuated as it propagates through deeper LLM layers, potentially limiting the model's ability to maintain effective visual grounding for navigation decision-making. Moreover, visual information also exhibits heterogeneous effects across different LLM layers. Strengthening visual representations at certain layers can enhance navigation reasoning, whereas reinforcing them at other layers may introduce perturbative signals that impair decision-making. Motivated by these observations, we propose Layer-Adaptive Visual Reinforcement (LAVR), a training-light mechanism that adaptively reconditions visual representations across multimodal decoder layers. Specifically, LAVR first evaluates each decoder layer in terms of both its need for visual reinforcement and the corresponding injection strength. A shared gated adapter then combines these signals to select a compact set of well-spaced layers. It further aligns the encoded visual features with the layer-wise hidden representations and adaptively integrates them into the selected layers. Experiments across multiple VLN architectures show that LAVR consistently improves closed-loop navigation performance and exhibits strong transferability, highlighting the importance of modeling layer-wise visual demands for more adaptive and effective visual information utilization in embodied decision-making. Code will be released to ensure full reproducibility.

Introduction

Vision-Language Navigation (VLN) is a fundamental task in embodied intelligence (Anderson et al. 2018b), requiring agents to follow natural-language instructions and make long-horizon navigation decisions from sequential visual observations. Existing VLN methods improve navigation decision-making by modeling visual observations and environmental context from different perspectives (Gu et al. 2022). One line of work (Zhang et al. 2024, 2025a; Qiao et al. 2022) directly reasons over 2D egocentric observations, projecting visual features into the language embedding space and jointly modeling them with instructions, trajectory histories, and candidate actions for action prediction. Some other works (Chen et al. 2022; Wang et al. 2023, 2021) augment these observations with topological graphs, online maps, or external memory to maintain global environmental context and support long-horizon planning. More recently, geometry-aware methods (Liu, Wang, and Yang 2024; Liu et al. 2023; Yang et al. 2026) explicitly reconstruct spatial structures using 3D scene representations or bird's-eye-view (BEV) maps, providing stronger spatial priors for navigation.

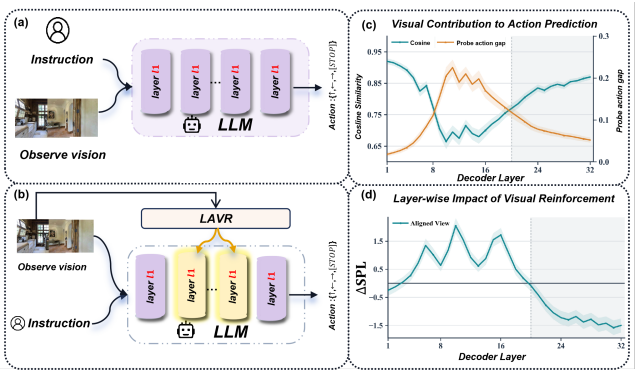


Figure 1: **Motivation of LAVR.** (a) A conventional VLN model processes visual observations and instructions through the multimodal decoder for action prediction. (b) LAVR adaptively reinforces visual representations at selected decoder layers. (c) Visual Contribution to Action Prediction analyzes how visual information contributes to navigation decisions across decoder depth. (d) Layer-wise Impact of Visual Reinforcement evaluates the navigation performance change caused by reinforcing visual information at different layers.

Despite these advances, existing studies have primarily focused on improving the visual observations provided to VLN agents, such as enhancing egocentric representations, incorporating global environmental context, or constructing explicit spatial structures. In other words, these methods mainly operate on the *observation vision* component in Figure 1(a), enriching or refining the visual information before it is fed into the multimodal decoder. In contrast, considerably less

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attention has been paid to what happens after these visual representations enter the decoder—particularly how they are preserved, transformed, and utilized throughout the internal decision-making process (Zhang et al. 2025b; Sun et al. 2025; Zhu and Feng 2026). Consequently, it remains unclear whether the visual evidence available at the input can be effectively retained and translated into reliable navigation actions.

In this paper, we analyze the internal evolution of visual information in multimodal decoders from a layer-wise perspective, examining how visual representations propagate through successive decoder layers and how their effects on navigation decision-making differ across layers. Our empirical analysis reveals that, first, visual information is progressively attenuated as it propagates through deeper LLM layers. As shown in Figure 1(c), visual information remains relatively prominent in the early layers, where the model progressively integrates visual observations with language instructions and navigation context. However, its strength decreases substantially in the deeper layers and remains weak near the output stage. This pattern indicates that the visual evidence encoded at the input is not consistently maintained throughout the decoding process, causing the final navigation decision to rely less on the observed environment. Second, the impact of visual information exhibits pronounced layer-wise heterogeneity. As illustrated in Figure 1(d), strengthening visual information at certain layers improves navigation performance, whereas the same intervention at other layers instead leads to performance degradation. Moreover, these beneficial and detrimental effects are not randomly scattered across the decoder but exhibit a structured layer-wise pattern, indicating that the utility of visual information depends strongly on the functional role of different decoder stages.

Motivated by these observations, we propose Layer-Adaptive Visual Reinforcement (LAVR), a training-light framework that adaptively reinforces visual representations across multimodal decoder layers to preserve effective visual information throughout navigation decision-making. LAVR comprises three key components. First, Visual Necessity Assessment (VNA) assigns each decoder layer a learnable selection score to estimate whether visual reinforcement is required under a fixed layer budget. Second, Injection Strength Estimation (ISE) measures the discrepancy between the projected visual features and the current layer-wise visual representations, providing a normalized estimate of the required reinforcement strength. Finally, the Gated Visual Adapter (GVA) jointly leverages these two signals to select a compact set of well-spaced layers. It aligns the encoded visual features with the corresponding hidden space and integrates them through a gated residual connection. Together with a two-stage training scheme, LAVR enables targeted visual reinforcement while reducing redundant interventions and perturbations to late-stage navigation decisions.

Our main contributions are summarized as follows:

- We conduct a layer-wise analysis of visual information propagation in multimodal VLN decoders and reveal two key characteristics: visual information is progressively attenuated toward deeper layers, while its contribution to navigation decision-making varies substantially across layers and exhibits a structured layer-wise pattern.
- We propose **Layer-Adaptive Visual Reinforcement (LAVR)**, a training-light framework that assesses layer-wise reinforcement demand and selectively injects visual information into decoder layers requiring additional visual support.
- Extensive experiments across multiple VLN benchmarks and model architectures demonstrate that LAVR consistently improves navigation performance and robustness over baselines, highlighting the importance of adapting visual reinforcement to the heterogeneous demands of different layers for reliable navigation decision-making.

Vision-Language Navigation

Vision-and-Language Navigation (VLN) requires embodied agents to ground natural-language instructions in visual observations and navigate previously unseen environments. Early discrete VLN methods formulated navigation as recurrent sequence prediction over predefined viewpoints connected by navigation graphs, simplifying control but relying on oracle graph connectivity (Anderson et al. 2018b). Vision-and-Language Navigation in Continuous Environments (VLN-CE) removes this assumption and requires agents to execute low-level actions in continuous 3D environments (Krantz et al. 2020). Subsequent waypoint-based and hierarchical approaches bridge instruction-level planning and physical control through waypoint prediction and online topological mapping (Krantz et al. 2021; An et al. 2025).

VLN models later shifted toward pretrained Transformer architectures. PREVALENT introduced large-scale vision–language–action pretraining for VLN (Hao et al. 2020), while VLN-BERT and HAMT improved cross-modal grounding and history-aware reasoning (Hong et al. 2021; Chen et al. 2021). DUET, GridMM, and VER represent navigation history using topological graphs, grid-based maps, and volumetric memories, respectively, to support global planning (Chen et al. 2022; Wang et al. 2023; Liu, Wang, and Yang 2024). More recent foundation-model-based agents jointly model visual observations, language instructions, navigation histories, and actions within shared generative backbones (Zhang et al. 2024, 2025a; Wei et al. 2025).

Layer-wise Sensitivity in Multimodal Decoders

Transformer layers do not contribute uniformly to model computation. ShortGPT estimates block importance using input–output similarity and prunes low-impact layers (Men et al. 2025), whereas LISA and MISA perform selective optimization based on layer- or module-wise importance to reduce memory costs (Pan et al. 2024; Liu et al. 2025). These studies challenge uniform processing across depth but mainly target language-model compression or memory-efficient optimization.

Visual information also evolves non-uniformly across multimodal decoder layers. A recent information-flow study identifies distinct stages of visual information transfer, cross-modal integration, and prediction formation across decoder depth (Zhang et al. 2025b). ShortV quantifies the contribution of layer transformations to visual tokens and freezes

visual-token updates in low-contribution layers (Yuan et al. 2025), while RedundancyLens reveals structured redundancy in visual-token processing within decoder-only MLLMs (Li et al. 2025). Together, these studies demonstrate that visual-token computation is strongly depth-dependent.

Unlike prior work that focuses on layer pruning, computation skipping, or memory-efficient optimization (Men et al. 2025; Pan et al. 2024; Liu et al. 2025; Yuan et al. 2025; Li et al. 2025), we examine how decoder depth governs navigation models' receptiveness to visual reconditioning. LAVR selectively reuses current-observation features at responsive layers, strengthening the observation-to-decision pathway without dense decoder-wide intervention.

Motivating Observations

To understand how visual information evolves and functions within multimodal decoders, we conduct a layer-wise empirical analysis and identify two key phenomena: **Visual Information Attenuation**, where visual signals progressively weaken toward deeper layers, and **Layer-wise Visual Impact**, where the contribution of visual information varies substantially across decoder layers and exhibits a structured layer-wise pattern.

Visual Information Attenuation

To examine whether visual evidence is consistently preserved throughout the multimodal decoder, we measure the layer-wise sensitivity of hidden representations and navigation decisions to visual perturbations. Given an original visual observation v, we construct a perturbed counterpart $\tilde{v}$ while keeping the instruction x, navigation history, and candidate actions unchanged.

Let $\mathbf{h}_{l,i}(v, x) \in \mathbb{R}^d$ denote the hidden state of the i-th token at decoder layer l under the original visual input, and let $\mathbf{h}_{l,i}(\tilde{v}, x)$ denote its counterpart under the perturbed input. We first measure the representation-level sensitivity to visual perturbation using the average cosine similarity:

$$C_l^{\text{hid}} = \frac{1}{N} \sum_{i=1}^{N} \cos\left(\mathbf{h}_{l,i}(v, x), \mathbf{h}_{l,i}(\tilde{v}, x)\right). \tag{1}$$

where N denotes the number of visual tokens.

To further examine whether the retained visual information influences navigation decisions, we attach an action probe $g_l(\cdot)$ to the hidden representation of each decoder layer. The probe maps the layer-wise representation to an action distribution, denoted by $\pi_l(\cdot \mid v, x)$ for the original visual observation and $\pi_l(\cdot \mid \tilde{v}, x)$ for its perturbed counterpart. We then quantify the decision-level sensitivity using the KL divergence:

$$S_l^{\text{act}} = D_{\text{KL}}\left(\pi_l(\cdot \mid v, x) \,\|\, \pi_l(\cdot \mid \tilde{v}, x)\right). \tag{2}$$

As shown in Figure 1(b), although visual perturbations induce noticeable changes in both hidden representations and action distributions in the early decoder layers, these differences gradually diminish toward the deeper layers. The hidden representations under v and $\tilde{v}$ become increasingly similar, accompanied by a substantial reduction in the action-distribution gap.

This indicates that visual information is progressively attenuated as it propagates through the decoder, leaving the later stages less responsive to changes in the observed environment.

Layer-wise Visual Impact

To characterize the layer-wise impact of visual information, we inject the encoded visual features into one decoder layer at a time and measure the resulting SPL change relative to the original model without additional visual injection. This controlled intervention isolates the contribution of visual information at each decoder depth while leaving the remaining model architecture and parameters unchanged.

As shown in Figure 1(c), the resulting performance gains are not smoothly distributed across decoder depth, nor do they follow a simple monotonic trend. Instead, several distinct peaks emerge in the early and intermediate layers, with a small subset of layers producing substantially larger SPL improvements than their neighboring layers. Such a structured multi-peak pattern suggests that different decoder stages differ markedly in their ability to exploit additional visual information, and that visually beneficial layers are concentrated at specific depths rather than uniformly distributed throughout the network. After approximately the 20th layer, this pattern changes sharply: the gains rapidly diminish, and visual injection consistently reduces SPL below the baseline. This transition indicates that the role of visual information is strongly layer-dependent. While selected early and intermediate layers can effectively incorporate additional visual evidence to support navigation reasoning, intervention in later layers increasingly disrupts the decision representations already formed by the decoder.

These observations motivate a layer-adaptive strategy that identifies and reinforces only the layers with genuine visual demand, rather than applying visual injection uniformly across the network.

Method

Motivated by the progressive attenuation and heterogeneous layer-wise effects of visual information in multimodal decoders, we propose **Layer-Adaptive Visual Reinforcement (LAVR)**, a training-light framework that adaptively reinforces visual representations according to the needs of individual decoder layers. By jointly modeling layer-wise reinforcement necessity and injection strength, LAVR enables targeted visual enhancement while reducing unnecessary perturbations to decision-making. In this section, we first present Visual Necessity Assessment and Injection Strength Estimation, which characterize whether each layer requires visual reinforcement and to what extent. We then introduce the Gated Visual Adapter, which integrates these two signals to identify the target layers, align the encoded visual features with the corresponding hidden representations, and perform adaptive visual integration. Finally, we describe the two-stage training procedure and the associated learning objectives.

Visual Necessity Assessment

To determine whether each decoder layer requires additional visual information, we associate the l-th layer with a learnable

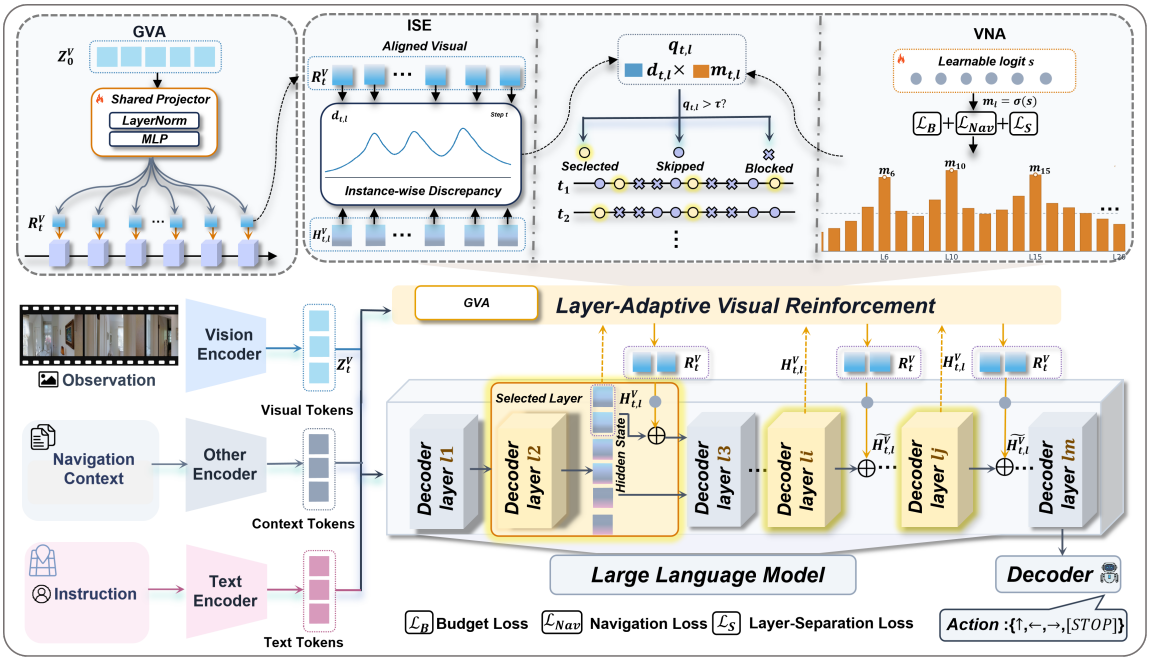


Figure 2: Overview of LAVR. Visual Necessity Assessment assigns each decoder layer a learnable score to estimate whether visual reinforcement is required under a fixed layer budget. Then, Injection Strength Estimation measures the discrepancy between the projected visual features and the layer-wise visual representations. Finally, the Gated Visual Adapter jointly leverages these two signals to select well-spaced target layers and integrate aligned visual features through gated residual connections.

scalar parameter s_l and map it into the interval $[0, 1]$ through a sigmoid function:

$$m_l = \frac{1}{1 + \exp(-s_l)}, \qquad l = 1, \ldots, L. \tag{3}$$

where m_l denotes the layer selection score of the l-th decoder layer. A larger m_l indicates a higher necessity for visual reinforcement, whereas a smaller value suggests that the layer can retain its original hidden representation without additional visual input.

The collection of scores $\mathbf{m} = [m_1, \ldots, m_L]$ provides a differentiable estimate of visual reinforcement necessity across decoder depth. Our motivating analysis shows that the benefits of visual reinforcement are concentrated in a small number of decoder layers, forming several distinct performance peaks rather than being uniformly distributed across depth. Reinforcing too many layers may therefore introduce redundant or perturbative visual signals. To reflect this sparse layer-wise pattern, we impose a global layer budget B on the selection scores, limiting the number of layers considered for reinforcement. The corresponding budget constraint is incorporated into the learning objective described later.

Injection Strength Estimation

While Visual Necessity Assessment estimates whether a decoder layer requires visual reinforcement, different layers may require different amounts of visual information. We therefore estimate the injection strength by measuring how far the current layer-wise visual representation deviates from the original encoded visual features.

Let $\mathbf{V} \in \mathbb{R}^{N_v \times d_v}$ denote the visual tokens produced by the visual encoder, and let $\mathbf{H}_l^V \in \mathbb{R}^{N_v \times d}$ denote the hidden states at the corresponding visual-token positions before the l-th decoder layer. Since these representations may lie in different feature spaces, we first project the encoded visual tokens into the decoder hidden space using a projector shared across all layers:

$$\mathbf{R}^V = \mathcal{P}_{\text{sh}}(\mathbf{V}). \tag{4}$$

where $\mathcal{P}_{\text{sh}}$ denotes the shared visual projector.

We then quantify their discrepancy using a normalized Frobenius distance:

$$d_l = \frac{\left\|\mathbf{R}^V - \mathbf{H}_l^V\right\|_F}{\left\|\mathbf{R}^V\right\|_F + \left\|\mathbf{H}_l^V\right\|_F + \epsilon}, \qquad d_l \in [0, 1). \tag{5}$$

where ϵ is a small constant for numerical stability. The resulting score d_l serves as the injection strength indicator of layer l. A larger value indicates that the current layer representation deviates more substantially from the original visual features and therefore requires stronger visual reinforcement, whereas a smaller value suggests that the visual information is relatively well preserved.

Gated Visual Adapter

Given the layer-wise necessity score m_l and injection strength indicator d_l, we jointly consider whether a layer requires visual reinforcement and how strongly its current representation deviates from the original visual features. Specifically, we define the layer selection score as

$$q_l = m_l \cdot d_l. \tag{6}$$

where a larger q_l indicates that the l-th layer exhibits both a stronger necessity for visual reinforcement and a larger visual representation discrepancy. A layer is treated as a candidate for visual injection when its selection score exceeds a predefined threshold τ.

Our layer-wise analysis further shows that beneficial visual reinforcement emerges at several separated performance peaks rather than in densely adjacent layers. This suggests that, after visual information is introduced at one layer, the subsequent decoder layers require sufficient depth to absorb, transform, and integrate the reinforced representation into the ongoing reasoning process. Re-injecting visual features too soon may interrupt this transition and introduce redundant or perturbative signals. We therefore impose a minimum layer gap G during layer selection. Specifically, once layer l is selected, the subsequent G layers are excluded from consideration, allowing the reinforced visual information to propagate and be assimilated before another injection is performed. Moreover, our motivating experiments show that visual reinforcement consistently yields negative gains beyond the 20-th decoder layer. As these deeper layers are closer to the output head and increasingly responsible for consolidating representations into final navigation decisions, introducing additional visual signals at this stage may disrupt the established decision process and perturb action prediction. We therefore restrict the candidate set to the first 20 layers.

Accordingly, the binary selection indicator is defined as

$$z_l = \mathbb{I}\left[q_l > \tau \ \wedge \ l \leq 20 \ \wedge \ l \notin \mathcal{E}\right]. \tag{7}$$

where $\mathcal{E}$ denotes the set of layers excluded by the gap constraint. Once layer l is selected, the exclusion set is updated as

$$\mathcal{E} \leftarrow \mathcal{E} \cup \{l+1, \ldots, \min(l+G, L)\}. \tag{8}$$

For each selected layer, the shared projector produces an aligned visual representation $\mathbf{R}^V$, which is incorporated into the hidden states at the corresponding visual-token positions through a gated residual connection:

$$\widetilde{\mathbf{H}}_l^V = \mathbf{H}_l^V + z_l \, m_l \, \mathbf{R}^V. \tag{9}$$

where $\widetilde{\mathbf{H}}_l^V$ denotes the reinforced visual hidden states. Here, z_l determines whether visual reinforcement is applied, while m_l controls the contribution of the aligned visual features. For unselected layers, $z_l = 0$, and the original hidden states remain unchanged. In this way, the adapter jointly leverages visual necessity and representation discrepancy for layer selection, while performing adaptive visual integration at the selected layers.

Learning Objectives

The overall training objective consists of the original navigation loss and two regularization terms that govern the distribution of visual reinforcement across decoder layers. Specifically, we define

$$\mathcal{L} = \mathcal{L}_{\text{nav}} + \lambda_{\text{b}} \left(\sum_{l=1}^{L} m_l - B \right)^2 + \lambda_{\text{s}} \sum_{d=1}^{G} \sum_{l=1}^{L-d} m_l m_{l+d}. \tag{10}$$

where $\mathcal{L}_{\text{nav}}$ denotes the original navigation objective, and λ_{b} and λ_{s} control the strengths of the budget and separation regularization terms, respectively.

The first regularization term imposes a global reinforcement budget B on the layer selection scores. Since $m_l \in (0, 1)$ provides a soft estimate of the reinforcement necessity of layer l, constraining their sum encourages the model to concentrate high selection scores on a limited number of layers rather than distributing visual reinforcement uniformly throughout the decoder. This is consistent with our motivating observation that performance gains are concentrated around only a few distinct layer-wise peaks.

The second term performs neighborhood suppression over a layer gap of G. It penalizes pairs of layers within a distance of at most G when both receive high selection scores. This encourages the learned reinforcement locations to remain sufficiently separated, allowing newly injected visual information to propagate, transform, and integrate through several decoder layers before another reinforcement is introduced. Together, these objectives promote a compact and well-spaced layer selection pattern while preserving the original navigation capability.

Training Scheme

We adopt a two-stage training scheme to stabilize the optimization of LAVR. In the first stage, we freeze the backbone model and optimize only the shared projector. Meanwhile, the layer selection scores are fixed to the same constant value across all decoder layers, such that visual reinforcement is applied in a uniform manner without introducing layer-specific preferences. This stage serves as a warm-up process, allowing the projector to learn a stable alignment between the encoded visual features and the hidden representation space of the multimodal decoder. It also enables the backbone to gradually adapt to the additional visual signals before layer-adaptive reinforcement is introduced, thereby reducing optimization instability caused by simultaneously learning feature alignment and layer selection.

In the second stage, we activate the learnable layer selection scores and jointly optimize them with the shared projector. The model then learns to differentiate the visual reinforcement necessity across decoder depth, while the injection strength is estimated from the discrepancy between the projected visual features and the corresponding layer-wise hidden representations. These two signals are combined by the Gated Visual Adapter to determine the target layers and perform adaptive visual integration. The layer budget and separation constraints are further incorporated into the learning objective to encourage a compact and well-spaced reinforcement pattern.

Experiments

Experimental Setup

Datasets. We evaluate LAVR on R2R-CE (Krantz et al. 2020) and RxR-CE (Ku et al. 2020; Krantz et al. 2020), two continuous-environment VLN benchmarks built from Matterport3D scenes (Chang et al. 2017). R2R-CE converts English Room-to-Room routes to continuous control, whereas

Host	Condition	R2R-CE val-unseen				RxR-CE val-unseen			
		NE↓	OSR↑	SR↑	SPL↑	NE↓	OSR↑	SR↑	SPL↑
Uni-NaVid† (Zhang et al. 2025a)	Host	4.96	57.4	51.8	47.7	6.71	60.8	48.1	42.6
	+ LAVR	4.88 (−0.08)	58.7 (+1.3)	54.1 (+2.3)	49.8 (+2.1)	6.38 (−0.33)	61.8 (+1.0)	49.6 (+1.5)	44.0 (+1.4)
StreamVLN (Wei et al. 2025)	Host	4.98	64.2	56.9	51.9	6.22	–	52.9	46.0
	+ LAVR	4.64 (−0.34)	65.6 (+1.4)	59.6 (+2.7)	54.6 (+2.7)	5.70 (−0.52)	–	54.6 (+1.7)	47.6 (+1.6)
GA-VLN† (Yang et al. 2026)	Host	4.836	67.21	60.63	55.21	5.88	67.0	55.4	45.2
	+ LAVR	4.301 (−0.535)	69.74 (+2.53)	64.05 (+3.42)	58.68 (+3.47)	5.42 (−0.46)	68.1 (+1.1)	56.9 (+1.5)	47.0 (+1.8)

Table 1: Navigation performance on R2R-CE and RxR-CE val-unseen. Values in parentheses indicate the absolute improvements over the corresponding host model. Results marked with † are obtained from our checkpoints.

RxR-CE contains longer trajectories and multilingual instructions. We follow the standard data splits and reserve val-unseen exclusively for evaluation.

Metrics. We report Navigation Error (NE), Oracle Success Rate (OSR), Success Rate (SR), and Success weighted by Path Length (SPL) (Anderson et al. 2018b,a). An episode succeeds when the agent stops within 3 meters of the goal (Krantz et al. 2020). NE measures final geodesic error, OSR records whether any visited position enters the success region, and SPL discounts successful episodes with inefficient trajectories. Lower NE and higher OSR, SR, and SPL indicate better navigation.

Baselines and host agents. The comparison includes modular planners, 3D end-to-end agents, and image-based MLLM agents. We additionally integrate LAVR into Uni-NaVid (Zhang et al. 2025a), StreamVLN (Wei et al. 2025), and GA-VLN (Yang et al. 2026), covering different host designs and DAgger regimes (Ross, Gordon, and Bagnell 2011). Each host retains its original observations, action space, navigation supervision, and evaluation code; LAVR uses no additional navigation data.

Implementation details. The backbone LLM and visual encoder remain frozen throughout training. In Stage I, layer selection is disabled, all candidate layers use a fixed necessity score of 0.15, and dense visual reinforcement is applied while training only the shared projector for 3,000 updates. In Stage II, the necessity scores are initialized to 0.15 and jointly optimized with the projector for another 1,000 updates under the full objective. Hard thresholding and layer exclusion are applied only during inference. We use AdamW (Loshchilov and Hutter 2017) with learning rates of 1×10^{-4} for the projector and 5×10^{-3} for the necessity scores.

By default, only the first 20 decoder layers are considered for reinforcement. We set the layer budget to $B = 3$, the exclusion width to $G = 4$, the selection threshold to $\tau = 0.15$, and the regularization coefficients to $\lambda_b = 0.1$ and $\lambda_s = 0.05$. Hyperparameters are tuned using only the training and val-seen splits. Unless otherwise specified, ablations are conducted on our reproduced GA-VLN checkpoint on R2R-CE val-unseen, and results are averaged over three runs.

Main Results

As shown in Table 1, LAVR consistently improves navigation performance across all three host architectures. On R2R-CE val-unseen, it yields SR gains of 2.3, 2.7, and 3.42 points for Uni-NaVid, StreamVLN, and GA-VLN, respectively, together with corresponding SPL improvements of 2.1, 2.7, and 3.47 points. Similar gains are observed on RxR-CE val-unseen, where LAVR improves both SR and SPL for every evaluated host while also reducing NE. The largest improvements are obtained with GA-VLN, indicating that LAVR can effectively complement stronger navigation backbones. Overall, these results demonstrate that layer-adaptive visual reinforcement provides consistent and transferable benefits across architectures and datasets.

Ablation Studies

Effectiveness of Layer Selection. To validate the effectiveness of the layers identified by LAVR, we keep the selected positions fixed and perturb the injected visual features by either shuffling the visual-token order or adding Gaussian noise. The results are shown in Table 2. Both variants still outperform the no-injection baseline, indicating that the selected layers are inherently suitable for receiving additional visual information. Injecting the original current-view features achieves the best performance, further showing that preserving the semantic and structural integrity of visual representations is essential for fully realizing the benefit of visual reinforcement.

Signal	NE↓	OSR↑	SR↑	SPL↑
None	4.836	67.21	60.63	55.21
Gaussian noise	4.447	68.16	61.61	56.70
Shuffled view	4.482	68.91	61.92	56.78
Current view	**4.301**	**69.74**	**64.05**	**58.68**

Table 2: Effectiveness of the selected reinforcement layers under different injected signals.

Ablation on Visual Reinforcement Signals. We ablate the two layer-wise signals used by LAVR: the visual necessity score m_l and the injection strength indicator d_l. As shown in Table 3, using either signal alone already improves over the model without visual reinforcement, confirming that both provide useful guidance for identifying beneficial intervention layers. VNA alone selects a fixed number of layers based only on the learned necessity scores, whereas ISE alone relies solely on the representation discrepancy. Combining the

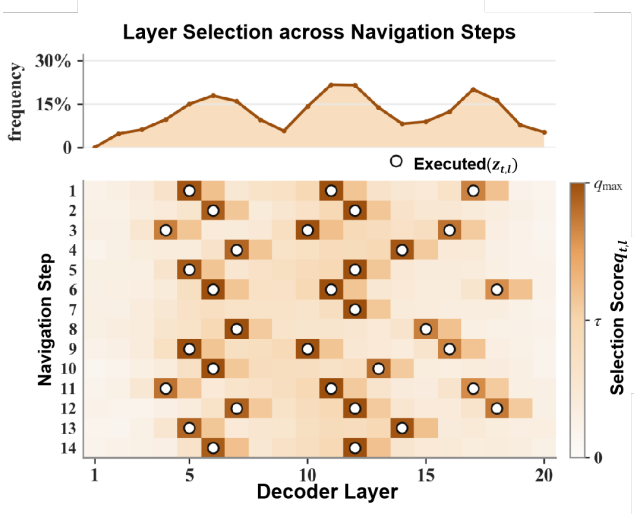


Figure 3: Layer Selection across Navigation Steps. The top panel shows the selection frequency of each candidate decoder layer, while the bottom panel visualizes the layer selection scores across navigation steps. White circles indicate the layers selected for visual reinforcement.

two signals in LAVR achieves the best SR and SPL compared with each variant. This demonstrates that m_l and d_l provide complementary information: the former captures whether a layer requires reinforcement, while the latter reflects the amount of visual information that should be introduced. Moreover, Figure 3 shows that the selected reinforcement layers vary across navigation steps and remain concentrated around several separated regions of the decoder. This pattern is consistent with our earlier observation that the benefit of visual reinforcement is highly heterogeneous across depth and emerges as several distinct layer-wise peaks rather than a uniform trend. Overall, these results show that jointly modeling reinforcement necessity and injection strength yields a sparse, state-dependent layer selection pattern consistent with the observed layer-wise heterogeneity.

Variant	NE↓	OSR↑	SR↑	SPL↑
No reinforcement	4.836	67.21	60.63	55.21
w/o ISE	4.359	69.47	63.68	58.26
w/o VNA	4.512	68.74	62.70	57.24
LAVR	**4.301**	**69.74**	**64.05**	**58.68**

Table 3: Ablation of the visual necessity and injection strength signals. It is noted that the injection weight is fixed to 0.15 at the selected layers for the ISE-only variant.

Ablation on Training Scheme. We examine the effectiveness of the proposed two-stage optimization strategy and the associated regularization terms. As shown in Table 4, jointly optimizing all components in a single stage from initialization results in inferior performance, indicating that the projector warm-up not only facilitates stable feature alignment but also allows the model to adapt to the visual injection mechanism before layer-adaptive reinforcement is introduced. In addition, removing the budget term leads to a less sparse allocation of reinforcement across decoder depth, which is inconsistent with our observation that beneficial interventions are concentrated around only a few layer-wise peaks. Likewise, removing the separation term permits reinforcement layers to cluster within a narrow depth range, overlooking the need for several intermediate layers to propagate and assimilate newly injected visual information. The full objective achieves the best performance, showing that stable visual alignment, appropriate sparsity, and a well-separated reinforcement pattern are critical for effective layer-adaptive visual injection.

Variant	NE↓	OSR↑	SR↑	SPL↑
single-stage	4.512	68.75	62.71	57.20
w/o budget	4.437	69.10	63.18	57.74
w/o separation	4.370	69.42	63.61	58.20
LAVR	**4.301**	**69.74**	**64.05**	**58.68**

Table 4: Ablation study on the two-stage training scheme and regularization terms on R2R-CE val-unseen.

Computational Overhead and Additional Analysis. LAVR introduces only limited computational overhead, as the host model remains frozen and the additional trainable components consist primarily of a shared visual projector and a small set of layer-wise scalar parameters. During inference, visual reinforcement is applied to only a sparse subset of decoder layers, resulting in an inference-time overhead of merely $1.05\times$ relative to the host model. Further sensitivity analyses of the layer budget B, exclusion gap G, selection threshold τ, and regularization coefficients, along with additional implementation details, are provided in the Appendix.

Conclusion

In this paper, we revisit visual information utilization in VLN from the perspective of decoder depth. Through layer-wise analysis, we show that visual representations are not preserved uniformly as they propagate through multimodal decoders, and that reinforcing them produces markedly different effects across layers. To this end, we introduce Layer-Adaptive Visual Reinforcement (LAVR), which learns where visual information should be reinforced and how much reinforcement is required. By combining visual necessity assessment, injection strength estimation, and a gated visual adapter, LAVR concentrates reinforcement on a compact set of informative and well-separated decoder layers while avoiding disruptive interventions near the output stage. Experiments across multiple VLN architectures demonstrate consistent improvements in closed-loop navigation with limited trainable overhead. Our results highlight the importance of explicitly regulating visual information across decoder depth for more effective embodied decision-making.

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